
Masking Cannot Remove What Is Inside the Region of Interest: Location-Dependent Shortcut Mitigation in Medical Imaging

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Abstract

1 Medical image classifiers exploit acquisition artifacts—hair, sonographer calipers,
2 intestinal debris—that correlate with the diagnosis. The standard defence restricts
3 the model to the region of interest (ROI). We show that this defence fails for ar-
4 tifacts that lie *inside* the ROI, explain why, and evaluate what to do instead. In
5 pre-registered, leakage-controlled traps built from real artifacts in four diseases
6 and three imaging modalities, the benefit of ROI masking is always larger for out-
7 of-ROI than for in-ROI artifacts (crossover +0.15 to +0.37 reversed-test AUROC,
8 all Holm-significant), across frozen and fine-tuned backbones and even in the zero-
9 shot scores of vision–language foundation models. Masking becomes harmful
10 when it also removes disease context: in controlled overlap sweeps, with end-to-
11 end fine-tuning (−0.28), and on unaltered test distributions (thyroid: −0.10 on
12 shortcut-conflicting cases; two of three held-out dermoscopy hospitals). A linear-
13 Gaussian analysis accounts for these signs in closed form and, fitted only to each
14 model’s clean and correlated performance, matches its held-out reversed-test AU-
15 ROC (mean absolute error 0.039) and the location crossover across 14 cohort-
16 backbone pairs ($r = 0.997$). As a remedy for frozen foundation-model features
17 we propose mask-then-erase with a generic insertion library (U-MtE): synthetic
18 overlays that resemble no particular artifact are inserted into masked images and
19 the feature subspace they move is erased, requiring no artifact examples, artifact
20 masks or artifact labels—only the ROI mask that masking already uses. U-MtE
21 removes in-ROI shortcuts carried by distinct overlays (+0.13 to +0.22 over mask-
22 ing on three backbones), erasure beats augmentation with the same overlays, a
23 disease-protected variant prevents failure when the artifact resembles the pathol-
24 ogy, and text-prompted erasure—the obvious vision–language alternative—is a
25 near no-op. Where the shortcut is carried by correlates of the artifact, label-based
26 reweighting is required. Code, pre-registrations and a backbone-agnostic tool are
27 released.

1 Introduction

29 Deep classifiers for medical images rely on features that correlate with the label but are not evi-
30 dence of disease [Geirhos et al., 2020, DeGrave et al., 2021]: surgical markings and rulers change
31 melanoma predictions [Winkler et al., 2019, 2021], and pneumothorax classifiers detect the chest
32 drain that treats the condition [Oakden-Rayner et al., 2020, Jiménez-Sánchez et al., 2023]. The most
33 common defence is to restrict the classifier to the region of interest (ROI) by masking the back-
34 ground. Its rationale assumes that shortcuts live outside the ROI. Hair, calipers and debris do not

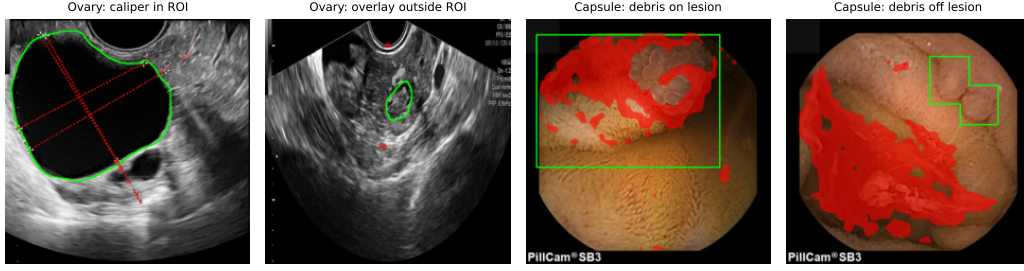


Figure 1: Real artifacts inside and outside the region of interest (green: ROI; red: artifact mask). Masking the background removes an out-of-ROI artifact (2nd, 4th) but keeps an in-ROI one (1st, 3rd) while deleting the context around it. Left: ovarian tumours (MMOTU); right: capsule endoscopy (SEE-AI).

respect that assumption: when the artifact overlaps the ROI, masking cannot remove it—it removes everything *else*.

We ask whether the value of a mitigation depends on where the artifact sits, which quantity governs that dependence, and how in-ROI shortcuts can be removed without localising the artifact. Our contributions:

1. **A location law for ROI masking**, established with pre-registered real-artifact traps in four diseases (dermoscopic hair; thyroid and ovarian sonographer calipers; capsule-endoscopy debris), replicated with three frozen backbones, two fine-tuned architectures, the zero-shot scores of vision–language models and on unaltered test data.
2. **A closed-form account** (Sec. 5) of when in-ROI masking harms, when erasure fails, and why group reweighting is location-invariant, which matches held-out results of real encoders in a pre-registered test.
3. **U-MtE**, a representation-level remedy for frozen foundation-model features that needs no artifact annotation (only the ROI mask), with evidence that erasure (not augmentation) is what works, a disease-protected variant, and a delineation of the regimes in which it does not apply.
4. **Rigour**: pre-registration before every experiment, Holm-corrected primary claims (12/16 supported), replication of an earlier pilot (73/88 numbers), and gaming-aware reporting.

2 Related work

Bissoto et al. [2020] showed that artifact debiasing in dermoscopy is harder than it appears and built trap sets; test-time selection [Bissoto et al., 2023] and generative repair [Jin et al., 2024] address specific artifacts. Hair occlusion degrades HAM10000 classifiers [Wang et al., 2024]; adding synthetic dark-corner artifacts to training improves classification [Pewton et al., 2024]; calipers are shortcuts even for fetal-ultrasound segmentation [Lin et al., 2024]; and classifiers on frozen foundation-model features show shortcut-like behaviour that pooled evaluation hides [Germani et al., 2026]. None treats the artifact’s location relative to the ROI as a variable. Group-robust training [Sagawa et al., 2020, Kirichenko et al., 2023, Liu et al., 2021] needs group labels or error sets; linear concept erasure [Ravfogel et al., 2020, Belrose et al., 2023] and its task-preserving form SPLINCE [Holstege et al., 2025] need concept labels; biased-prompt projection [Chuang et al., 2023] derives directions from text. Random erasing and occlusion augmentation [Zhong et al., 2020, Fong and Vedaldi, 2019] use overlays as training noise. We estimate an artifact subspace from synthetic *insertion* pairs and remove it only after masking.

3 Setup

Traps Let Y be the diagnosis and A the presence of an artifact. A trap trains with $P(A=1 \mid Y=1) = 0.9$, $P(A=1 \mid Y=0) = 0.1$ and tests on a correlated (same rates), a reversed (swapped)

Table 1: Real-artifact cohorts. Artifact masks: published (Wegley et al.), audited detector (95% precision on 40 images), or expert-trained probe (pixel accuracy 0.92).

Cohort	Modality	Images	Artifact (mask source)	ROI
ISIC 2019	dermoscopy	25,331	hair/ruler (published masks)	lesion
TN3K/TNCD	thyroid ultrasound	3,493	calipers (rule-based detector)	nodule
MMOTU	ovarian ultrasound	1,202	calipers/overlays (same detector)	tumour
SEE-AI	capsule endoscopy	5,481	debris/bubbles (patch-token probe)	lesion boxes
NIH ChestX-ray14	radiography	29,687	chest drains (image-level)	lungs

and a clean environment (artifact independent of the label: present in half of each class in real traps, absent in sweeps). The overlap $r = |\text{artifact} \cap \text{ROI}|/|\text{artifact}|$ defines Trap A (in-ROI, $r \geq 0.5$) and Trap B (out-of-ROI, $r < 0.1$); both share one artifact-free group, cells are matched on source and label, and splits are grouped by lesion/patient identifiers where published, otherwise by perceptual hashes and frame blocks (five seeds $\times$ five folds; a stricter embedding-based grouping is a sensitivity analysis). The primary metric is reversed-test AUROC; we always report correlated and clean AUROC, and summarise robustness by $\min(\text{reversed}, \text{correlated})$, which penalises methods that merely flip the shortcut. Uncertainty is a hierarchical paired bootstrap (10,000 replicates, clusters = seeds or folds).

Cohorts (Table 1; details in Appendix A.) Heads are linear on frozen DINOv2 ViT-B/14, MedSigLIP, DermLIP, RAD-DINO or ConvNeXt features; regularisation and threshold are chosen on clean validation data only. Controlled sweeps paste a synthetic artifact (ruler, caliper, debris, tube) at overlap $r \in \{0, 0.25, 0.5, 0.75, 1\}$.

4 Mask-then-erase with a generic insertion library

Removing an in-ROI artifact requires its pixels; inserting one does not. A procedural generator $\mathcal{G}$ (strokes, bands, patches, blobs, dashed lines, glyphs; random colour, opacity and position, half centred in the ROI) turns every masked training image $m(x_i)$ into a counterfactual $m(\mathcal{G}(x_i))$; the feature differences $\Delta_i = f(m(\mathcal{G}(x_i))) - f(m(x_i))$ are independent of Y . U-MtE erases their span,

$$U_k = \text{top-}k \text{ eigvecs}\left(\frac{1}{n} \sum_i \Delta_i \Delta_i^\top\right), \quad P(z) = z - U_k U_k^\top (z - \mu), \quad (1)$$

with k chosen without labels (90% held-out difference energy), and fits the head on $P(f(m(x)))$. The generator contains no ruler, hair, caliper or debris template. *Disease protection* erases $U'_k = \text{orth}((I - WW^\top)U_k)$, where W spans logistic label directions fitted on artifact-free training images, guaranteeing $w^\top P(z) = w^\top z$ for $w \in \text{span}(W)$. With image-level artifact labels the head is group-balanced (U-MtE_{bal}).

5 Why location matters

Let the disease block have class-separation SNR S and the artifact block SNR A per unit of presence. For the LDA-optimal linear head trained with (p_1, p_0) , the artifact's effective SNR is $\tilde{A} = (p_1 - p_0)^2 A / (1 + p_1(1 - p_1)A)$ and

$$\text{AUROC}_{\text{rev}} = \Phi\left(\frac{S - \tilde{A}}{\sqrt{2(S + \tilde{A})}}\right), \quad (2)$$

increasing in S and decreasing in $\tilde{A}$ (simulation agreement ≤ 0.009 AUROC; Appendix B). Masking maps $(S, \tilde{A}) \mapsto (S_m, \tilde{A}_m)$ with $\tilde{A}_m = 0$ out of the ROI and $\tilde{A}_m \geq \tilde{A}$ inside. Hence in-ROI masking harms exactly when it removes disease context ($S_m < S$); the crossover is positive for every S_m ; erasure sets $\tilde{A} \rightarrow 0$ but scales S by $1 - \rho^2$ (ρ : overlap of the erased subspace with the disease direction), so it fails for pathology-like artifacts unless protected; and group reweighting yields $\Phi(\sqrt{S/2})$ regardless of location (simulation in Appendix Fig. 5).

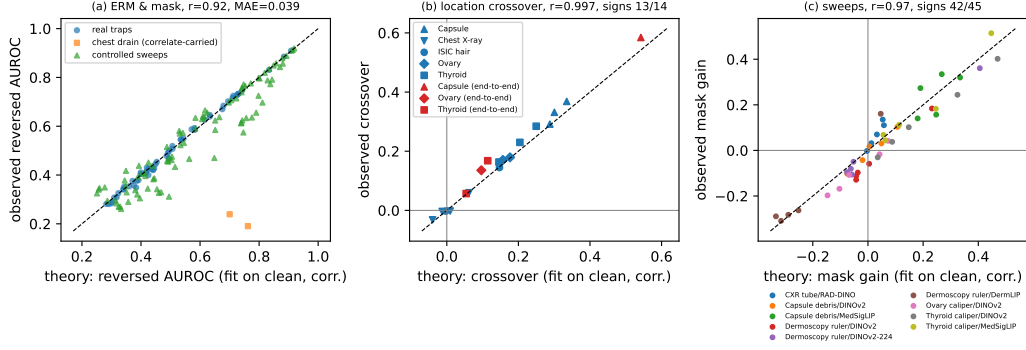


Figure 2: Pre-registered test of the closed form on real encoders: (S, A) fitted to each model’s clean and correlated AUROC, reversed-test results held out. (a) Reversed-test AUROC (ERM, masking). (b) Location crossover per cohort and backbone (red: end-to-end fine-tuned, outside the model’s assumptions). (c) Mask gain along controlled sweeps.

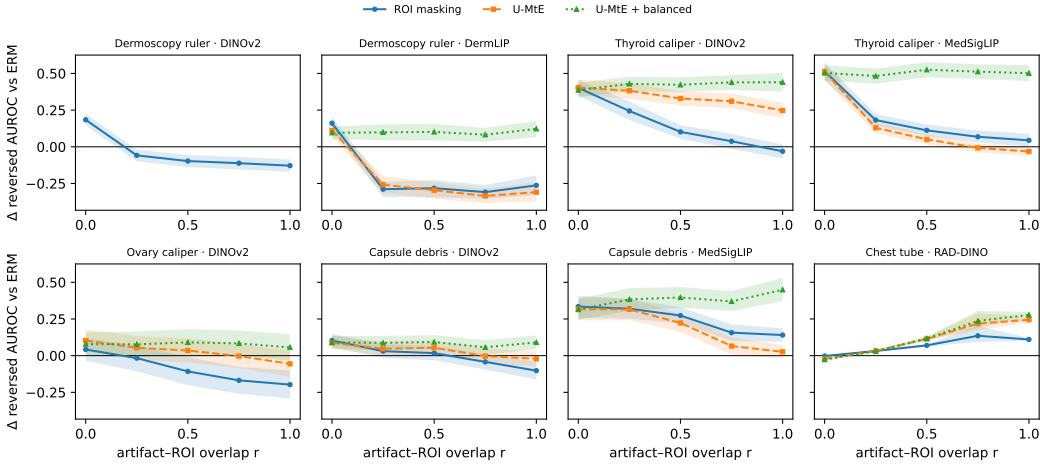


Figure 3: Controlled overlap sweeps: Δ reversed-test AUROC versus ERM as a synthetic artifact moves into the ROI (95% CIs). Masking (blue) declines with overlap except in chest radiography, where the encoder ignores extra-pulmonary tubes; U-MtE with group balancing (green) is flat.

103 **Test on real encoders** The same head gives the AUROC of any test composition (Appendix B). In
104 a pre-registered test we fitted (S, A) for every trained model to its clean and correlated AUROC only,
105 with no shared or tuned parameter, and compared the implied reversed-test AUROC with the held-
106 out observation (Fig. 2): mean absolute error 0.039 over 148 ERM and masking models ($r = 0.92$;
107 0.108 for a linear-symmetry heuristic). The implied location crossover agrees with the observed one
108 across 14 cohort–backbone pairs ($r = 0.997$, error 0.014, 13/14 signs; the miss is a chest-radiograph
109 cell at -0.002) and also for fine-tuned networks (4/4 signs). One sign miss means that, by our pre-
110 registered criterion, the model remains a qualitative account; it fails only for chest drains, whose
111 shortcut is carried by correlates that the clean and correlated environments do not expose.

112 6 Results

113 6.1 A location law for ROI masking

114 In all four real-artifact cohorts the crossover is positive and survives Holm correction (Table 2); it
115 replicates with MedSigLIP and ConvNeXt, with end-to-end fine-tuning (ResNet-50 capsule; ViT-S
116 thyroid), and in the zero-shot scores of MedSigLIP and DermLIP, i.e. before any task training. Mask-

Table 2: Masking gain (mask – ERM, reversed AUROC) is always larger for out-of-ROI than in-ROI artifacts. Crossover = gain(out) – gain(in), 95% CI. Zero-shot: no task training. Natural: shortcut-conflicting pairs of untouched test data.

Setting	Cohort (backbone)	Crossover [95% CI]
Real traps, frozen	ISIC hair (DINOv2)	+0.145 [+0.105, +0.181]
	Thyroid (DINOv2 / MedSigLIP / ConvNeXt)	+0.230 / +0.163 / +0.284
	Ovary (DINOv2)	+0.170 [+0.116, +0.224]
	Capsule (DINOv2 / MedSigLIP / ConvNeXt)	+0.368 / +0.332 / +0.292
Fine-tuned	Capsule (ResNet-50)	+0.584 [+0.469, +0.691]
	Thyroid (ViT-S)	+0.168 [+0.074, +0.265]
Zero-shot VLM	Thyroid / Capsule (MedSigLIP)	+0.103 / +0.049
	Ovary (MedSigLIP) / Hair (DermLIP)	+0.181 / +0.051
Natural, mask – ERM	Thyroid (official split)	−0.102 [−0.139, −0.066]
	ISIC held-out BCN / MSK	−0.016 / −0.056

Table 3: Unaltered data (no resampling): change in AUROC on shortcut-conflicting pairs. Thyroid: official TN3K split; ISIC: each hospital held out in turn; capsule: held-out frame blocks. 95% CIs.

Test set	mask – ERM	U-MtE – mask	U-MtE _{bal} – mask
Thyroid (official split)	−0.102 [−0.139, −0.066]	+0.051 [+0.033, +0.072]	+0.088 [+0.071, +0.108]
ISIC, BCN held out	−0.016 [−0.027, −0.005]	+0.003 [+0.000, +0.006]	+0.034 [+0.029, +0.039]
ISIC, MSK held out	−0.056 [−0.084, −0.029]	−0.008 [−0.015, −0.002]	+0.041 [+0.029, +0.053]
ISIC, HAM held out	+0.053 [+0.037, +0.068]	+0.023 [+0.017, +0.032]	+0.065 [+0.056, +0.075]
Capsule (held-out frames)	+0.021 [+0.000, +0.044]	+0.000 [−0.010, +0.009]	+0.014 [+0.003, +0.025]

ing becomes *harmful* for in-ROI artifacts when it also removes disease context: real dermoscopic hair (−0.037 [−0.057, −0.019]), controlled sweeps at full overlap (ovary −0.197 [−0.292, −0.100]; capsule −0.102 [−0.162, −0.040]; Fig. 3), fine-tuned capsule (−0.284 [−0.347, −0.226]) and on unaltered data, where masking lowers AUROC on shortcut-conflicting cases for thyroid and two of three held-out ISIC hospitals (Table 3); at the validation-fixed operating point it halves thyroid sensitivity (0.734 → 0.347; AUPRC, Brier score and calibration in Appendix Table 6). The claims are robust to the artifact-label definition (large-marker and strict-location variants) to a visual audit of the masks and to a stricter, embedding-based leakage grouping (11 of 12 primary estimates unchanged; Appendix E). Chest radiography is a boundary case: the encoder ignores extra-pulmonary tubes, so masking has nothing to remove outside the lungs.

Mechanism. Three explanations were tested separately in the controlled dermoscopy sweep (Appendix C). *Occlusion* is not the cause: with the artifact present but uncorrelated with the label, masking helps at every overlap. *Lost context* alone is not the cause: dilating the mask at 50% overlap increases harm as it restores artifact pixels, and at full overlap harm is flat although the artifact’s share of the visible pixels falls threefold. The governing quantity is artifact *retention*: masking keeps an in-ROI artifact, and the masked model’s counterfactual sensitivity to it roughly doubles.

6.2 Removing in-ROI shortcuts

Without any artifact example, artifact mask or label, U-MtE improves on masking where the artifact is a distinct overlay: thyroid +0.222 [+0.197, +0.251] (DINOv2), +0.204 [+0.174, +0.237] (MedSigLIP), +0.128 [+0.112, +0.145] (ConvNeXt); ovary +0.215 [+0.182, +0.246] (MedSigLIP); controlled sweeps +0.08 to +0.28; end-to-end (ResNet-50, thyroid) +0.143 [+0.099, +0.188]. The same overlays used as training augmentation gain at most +0.03; erasing their subspace gains +0.207 [+0.182, +0.235] more on thyroid. On capsule debris, which resembles erosion fibrin, unprotected erasure removes disease signal; disease protection removes the failure. With image-level artifact labels, U-MtE_{bal} is the most robust model on thyroid with all three backbones (Table 4; full values in Table 7). Where the shortcut is carried by correlates of the artifact (hair co-varies with body site, age and sex; a drain comes with a treated lung) no pixel- or insertion-based method helps much and DFR [Kirichenko et al., 2023] is best.

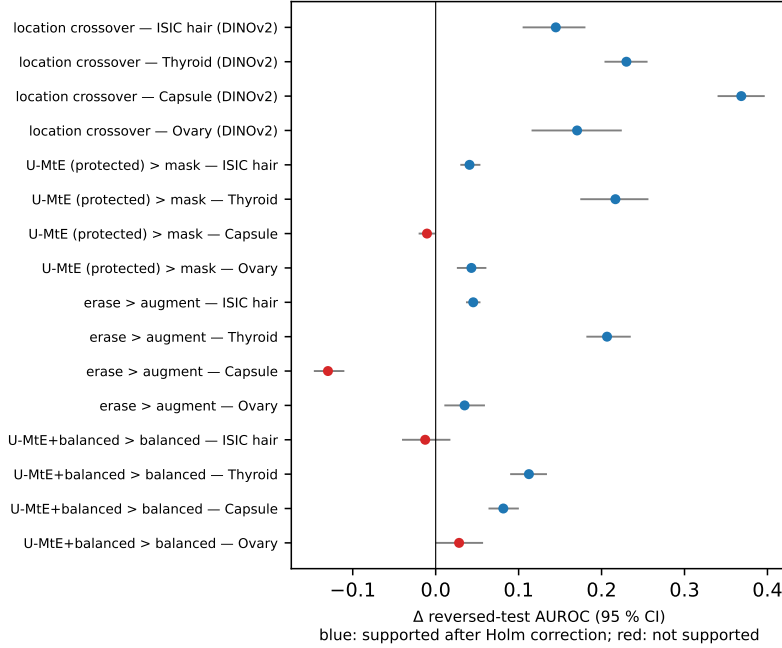


Figure 4: The 16 pre-registered primary tests with Holm correction (blue: supported, 12/16; red: not supported).

Table 4: In-ROI artifacts (Trap A): robustness = $\min(\text{reversed}, \text{correlated})$ AUROC, which penalises both shortcut use and shortcut flipping. Bold: best per column. Full reversed/correlated/clean values for all cohorts and backbones: Table 7.

Method	Needs	Hair DINOv2	Thyroid DINOv2	Thyroid MedSigLIP	Ovary DINOv2	Capsule DINOv2	Drain RAD-DINO
ROI masking	ROI	0.45	0.40	0.35	0.55	0.68	0.24
Mask + overlay augmentation	ROI	0.46	0.42	0.38	0.53	0.70	0.24
U-MtE (ours)	ROI	0.51	0.62	0.55	0.56	0.57	0.27
U-MtE, protected (ours)	ROI	0.49	0.62	0.61	0.59	0.67	0.27
Group-balanced	A	0.72	0.68	0.70	0.71	0.78	0.54
DFR	A	0.74	0.66	0.72	0.69	0.80	0.68
Mask + DFR	ROI + A	0.69	0.74	0.76	0.75	0.88	0.65
U-MtE + balanced (ours)	ROI + A	0.70	0.79	0.79	0.74	0.86	0.46

Baselines SPLINCE [Holstege et al., 2025], fitted with artifact labels, removes much disease signal when artifact and label are strongly associated (clean AUROC 0.56–0.78) and flips the shortcut on capsule; U-MtE_{bal} is more robust on thyroid, capsule and hair (0.788 vs 0.596; 0.857 vs 0.683; 0.704 vs 0.564). Text-prompted erasure (biased prompts, Chuang et al., 2023) is a near no-op ($\leq +0.008$ over masking): the direction along which a *described* artifact moves the text embedding is not the one along which the real artifact moves the image embedding. Oracle-mask LaMa inpainting [Suvorov et al., 2022] is strong when artifact masks exist; protected U-MtE matches it on thyroid without masks. Label-free, U-MtE beats JTT [Liu et al., 2021] on thyroid (+0.198) and ovary (+0.137). A validation-driven selector beats masking in all five cohorts but not DFR everywhere (Appendix F).

7 Discussion and limitations

What carries the shortcut decides the method Across six real and controlled settings three regimes emerge. (1) *Distinct overlays that carry the shortcut in their pixels* (sonographer calipers, synthetic rulers, debris and tubes): masking removes an out-of-ROI copy but leaves an in-ROI copy

intact, and U-MtE removes most of what masking leaves (thyroid +0.22, +0.20, +0.13 on three backbones; controlled sweeps +0.08 to +0.28). (2) *Artifacts that resemble the pathology* (capsule debris versus erosion fibrin): any subspace estimated from artifact appearance removes disease evidence; disease protection makes erasure safe, and with image-level artifact labels masking plus last-layer retraining is the most robust option. (3) *Artifacts whose shortcut is carried by correlates* (dermoscopic hair co-varies with body site, age and sex; a chest drain comes with a treated, re-expanded lung): removing even every artifact patch with an oracle mask gains little (+0.015 for hair), no pixel- or insertion-based method can close the gap, and label-based group methods (DFR, balancing, JTT on hair) are preferable. The oracle-removal ceiling is a cheap diagnostic of the regime.

Why erasure works where masking and patch removal do not In thyroid ultrasound, dropping every patch token that contains a caliper (oracle masks) before mean-pooling improves the reversed AUROC by only +0.047 over pooling all ROI tokens, whereas erasing the generic-overlay subspace from the class-token representation improves on masking by +0.22 (the two representations differ, so the comparison is indicative). Self-attention distributes the information about a salient overlay across the whole representation; an in-ROI artifact is therefore not contained in “its” pixels or patches, and a representation-level operation is needed to remove it.

Erasure, not augmentation The same overlays used as training augmentation gain at most +0.03 over masking in every cohort; used to estimate and erase a subspace they gain up to +0.21 more. Augmentation teaches invariance to the *generated* overlays; erasure removes the directions along which *any* overlay moves the representation, which transfers to real artifacts the generator never produced.

Reading reversed-environment gains A reversed-test gain can be manufactured by removing diagnosis-correlated signal together with the artifact: erasure fitted on unpaired image-level labels and post-hoc prevalence recalibration invert the correlated/reversed ordering. We report reversed, correlated and clean AUROC for every method and summarise robustness by the minimum of correlated and reversed AUROC.

Decision rule for practitioners (1) Measure where artifacts sit relative to the ROI; out-of-ROI artifacts are removed by masking. (2) For in-ROI artifacts without any artifact annotation, use protected U-MtE; it matches oracle-mask LaMa inpainting where the artifact is a distinct overlay. (3) With image-level artifact labels, combine masking or U-MtE with group balancing or last-layer retraining, choosing among them on a group-aware validation split; prefer label-based reweighting alone when the shortcut is carried by correlates of the artifact, which the oracle-removal ceiling (a handful of annotated masks) reveals.

Limitations (i) U-MtE is a representation-level intervention for frozen (foundation-model) features with linear heads, the setting in which it is evaluated throughout; end-to-end fine-tuning is a secondary check (the location law replicates, the U-MtE analogue works on thyroid with ResNet-50 but not with ViT-S), not a full sweep. (ii) Artifact masks for thyroid calipers and capsule debris come from an audited rule-based detector (precision/recall 0.86–0.89) and a patch-token probe (IoU 0.62); errors act as label noise on A and on the overlap strata. (iii) Hair and ruler share one annotation channel in the ISIC 2019 archive. (iv) No patient or video identifiers are published for TN3K, MMOTU or SEE-AI; leakage groups use perceptual hashes and frame blocks, and a stricter embedding-based grouping leaves 11 of 12 primary estimates unchanged (Sec. E.5). (v) Chest-drain labels for pneumothorax-negative radiographs come from a detector. (vi) The controlled chest-X-ray sweep did not show the sign reversal: the encoder ignored extra-pulmonary tubes. (vii) Reversed environments are constructed by subsampling; real deployment shifts are less extreme, and the clean results bound the cost without shift. (viii) The disease-protection step needs diagnosis labels of artifact-free images, and the balanced variants need image-level artifact labels. (ix) The linear-Gaussian account is fitted per model; it cannot anticipate a shortcut carried by correlates that the clean and correlated environments do not expose (chest drains).

Broader impact Shortcut audits that report only overall accuracy can certify masked models that still exploit in-ROI artifacts, and a mitigation that flips a shortcut can look like a success on a

211 reversed test. We recommend reporting artifact location relative to the ROI and performance on
212 shortcut-conflicting cases together with clean and correlated performance.

213 8 Conclusion

214 ROI masking is not neutral: it removes shortcuts outside the ROI and leaves—or amplifies—those
215 inside it. This holds across diseases, modalities, training regimes and foundation models, and a two-
216 parameter model fitted to clean and correlated performance matches the held-out results. Erasing
217 a subspace learned from generic synthetic overlays removes in-ROI shortcuts carried by distinct
218 artifacts without any annotation of the artifact.

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289 A Data

290 **Controlled dermoscopy cohort** ISIC 2018 Task 1/2 images with manual lesion masks and
291 melanoma labels from the union of the ISIC 2016 and 2017 tables (three images with conflicting
292 labels dropped); perceptual-hash (Hamming ≤ 8) and lesion grouping; 2,437 images for which a
293 65×19 px ruler (0.46% of a 518×518 image) can be placed at 0/25/50/75/100% overlap with the
294 lesion within 10 percentage points (train/val/test 1,849/371/371 before common support).

Real hair cohort (ISIC 2019) All 25,331 ISIC 2019 training images; hair/ruler masks from the archive of Wegley et al. [2026]; lesion masks manual for HAM10000 [Tschandl et al., 2018] and predicted elsewhere by a U-Net trained on ISIC 2018 Task 1. Two traps share one hair-free group (≤ 30 hair pixels at native resolution): Trap A contains hair with overlap $r = |\text{hair} \cap \text{lesion}|/|\text{hair}| \geq 0.5$, Trap B $r < 0.1$. Cells are subsampled to the hair-free source mix (HAM/BCN/MSK) within each label, giving Trap A 704/157/2,738/989 and Trap B 704/157/4,327/930 images (hair-free benign/melanoma, hair benign/melanoma); five seeds of five-fold group-safe cross-validation (leakage groups: lesion identifiers and pHash); 172 reversed-test melanomas per trap and seed.

Thyroid ultrasound TN3K images with TNCD benign/malignant labels (3,493 images; 1,210 malignant) and manual nodule masks. Sonographer calipers were segmented by a frozen rule-based detector (dotted measurement lines and ‘+’ ends; audit in Sec. 7). Trap A: calipers with $r \geq 0.5$ (1,150 images); Trap B: $r < 0.1$ (326); caliper-free: 1,801. No patient identifiers are published; leakage groups are pHash chains (stricter embedding-based groups in Sec. E.5).

Ovarian tumour ultrasound MMOTU OTU_2d [Zhao et al., 2022] (CC BY 4.0): 1,202 images with tumour masks, labelled benign cystic (chocolate cyst, serous cystadenoma, simple cyst) versus lesions with solid components (teratoma, theca-cell tumour, mucinous cystadenoma, high-grade serous carcinoma). The thyroid detector is applied unchanged: Trap A 341/276, Trap B 91/103, caliper-free 170/178 (benign-cystic/solid).

Capsule endoscopy SEE-AI frames [Yokote et al., 2023] (CC BY 4.0) containing only erosions (3,550) or only polyp-like lesions (1,931); the ROI is the union of the expert lesion boxes. Contamination (debris, bubbles, bile) is segmented by a linear probe on DINOv2 patch tokens trained on 2,160 expert-masked frames (held-out pixel accuracy 0.92, IoU 0.62; expert masks are used where available). Debris-free: $< 3\%$ of the frame; Trap A: debris $\geq 10\%$ with $r \geq 0.5$; Trap B: $r < 0.1$. Leakage groups: blocks of 100 consecutive frame numbers plus pHash duplicates.

Chest radiography NIH ChestX-ray14 [Wang et al., 2017]; lung ROIs from the TorchXRaiVision PSPNet [Cohen et al., 2022]. *Controlled cohort*: 1,906 pneumothorax images without a chest drain (NEATX annotations) and 5,718 *No Finding* images from other patients; a 100×12 px radio-opaque tube is placed at controlled overlap with the lungs (7,610 images in common support). *Real drains* (in-ROI only): 29,687 images; drains are labelled by NEATX for pneumothorax-positive images and by a drain detector (precision 0.95) for negatives; environments are built within view position (AP/PA); five patient-grouped folds.

B Why location matters: a linear-Gaussian account

Let a frozen encoder yield a disease block with class-separation signal-to-noise ratio S and an artifact block with SNR A per unit of artifact presence; training uses $P(A=1 | Y=1) = p_1$, $P(A=1 | Y=0) = p_0$ and the reversed test swaps them. For the LDA-optimal linear head the artifact’s effective training SNR is $\tilde{A} = (p_1 - p_0)^2 A / (1 + p_1(1 - p_1)A)$ and

$$\text{AUROC}_{\text{rev}}(S, \tilde{A}) = \Phi\left(\frac{S - \tilde{A}}{\sqrt{2(S + \tilde{A})}}\right), \quad (3)$$

increasing in S and decreasing in $\tilde{A}$ (verified by simulation to within 0.009 AUROC; Fig. 5). Masking maps $(S, \tilde{A}) \mapsto (S_m, \tilde{A}_m)$. Out of the ROI, $\tilde{A}_m = 0$; inside, $\tilde{A}_m \geq \tilde{A}$. Hence (i) in-ROI masking harms exactly when the mask removes disease-informative context ($S_m < S$) and helps only by denoising ($S_m > S$); (ii) for any S_m the benefit of masking is larger out of the ROI than inside it—the crossover is always positive; (iii) erasing a subspace that contains the artifact direction sets $\tilde{A} \rightarrow 0$ but scales S by $1 - \rho^2$, where ρ is its overlap with the disease direction, so erasure beats in-ROI masking unless the artifact resembles the pathology, which disease protection prevents; (iv) group balancing removes the artifact–label association and everything correlated with it, giving $\Phi(\sqrt{S/2})$ irrespective of location; (v) a shortcut carried by correlates that neither the mask nor pixel-level erasure touch can only be removed by label-based reweighting. Each consequence is borne out in Sections C–F.

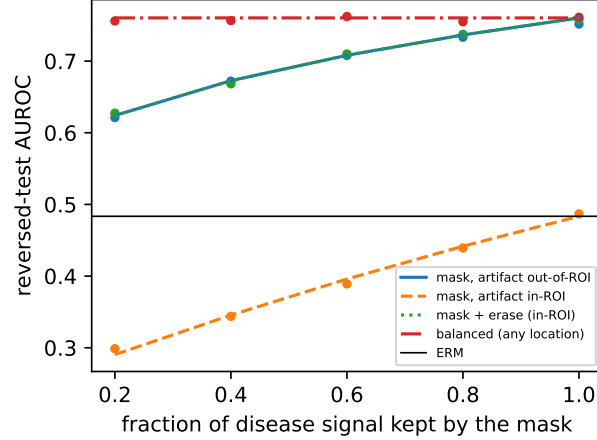


Figure 5: Reversed-test AUROC predicted by the linear-Gaussian model (lines) and simulated with logistic heads (points) as masking removes a growing share of disease signal ($S = 1$, $A = 2$, 90/10 training). Generated by `scripts/analysis/theory_sim.py`.

342 **Does the account match real encoders?** For any test composition $q_y = P(A=1 \mid Y=y)$ the
343 same head gives $\text{AUROC} = \Phi(\Delta / \sqrt{\sigma_1^2 + \sigma_0^2})$ with $\Delta = S + (p_1 - p_0)(q_1 - q_0)A / (1 + vA)$,
344 $\sigma_y^2 = S + w_a^2(1 + Aq_y(1 - q_y))$, $w_a = \sqrt{A}(p_1 - p_0) / (1 + vA)$ and $v = [p_1(1 - p_1) + p_0(1 - p_0)] / 2$.
345 In a pre-registered test we fitted (S, A) for every trained model to its clean and correlated AUROC
346 *only* and compared the implied reversed-test AUROC with the held-out observation; no parameter
347 is shared across models or tuned (Fig. 2). Over 148 ERM and masking models (real traps, chest
348 drains, controlled sweeps) the mean absolute error is 0.039 AUROC ($r = 0.92$), against 0.234 for
349 “reversed = clean” and 0.108 for a linear-symmetry heuristic (2-clean – correlated). The implied
350 location crossover agrees with the observed one across 14 cohort–backbone pairs ($r = 0.997$, mean
351 absolute error 0.014, 13/14 signs; the miss is a chest-radiograph cell with an observed crossover of
352 -0.002), the implied mask gains agree with the controlled sweeps ($r = 0.965$, 42/45 signs), and
353 the agreement extends to end-to-end fine-tuned networks (4/4 crossover signs), which violate the
354 model’s assumptions. Because our pre-registered criterion required every crossover sign to agree,
355 we still call the model a qualitative account. Its one clear failure is the chest-drain cohort (implied
356 0.70–0.76, observed 0.19–0.24): the clean and correlated environments do not expose a shortcut
357 carried by correlates of the drain, regime (v).

358 C Results: dermoscopy

359 C.1 Masking reverses with location (controlled)

360 With a synthetic ruler whose overlap with the lesion is swept from 0 to 100% (2,437 ISIC 2018
361 images, DINOv2 ViT-B/14 at 518 px, three seeds), ROI masking improves reversed-environment
362 AUROC over ERM by +0.184 [+0.146, +0.224] when the ruler lies outside the lesion and *low-*
363 *ers* it by -0.097 [$-0.138, -0.056$], -0.112 [$-0.155, -0.068$] and -0.129 [$-0.170, -0.087$] at 50,
364 75 and 100% overlap (location interaction +0.313 [+0.272, +0.353]; Table 5). The sign flip is
365 reproduced at 224 px (+0.361 at 0%; -0.095 at 100%) and under randomised ruler colour, opac-
366 ity, ticks and scale. At 0% overlap the masked model’s correlated and reversed AUROC coincide
367 (0.849/0.849): the shortcut is removed. At 100% the masked model’s correlated AUROC exceeds
368 ERM’s (0.961 vs 0.937): it uses the ruler *more*.

369 Insert-then-erase in this table is the artifact-specific precursor of U-MtE: its insertion pairs use tem-
370 plates of the artifact type (rulers) and it erases from unmasked features. U-MtE (Sec. 4) replaces the
371 templates by one generic overlay library for every modality and erases after masking; its dermoscopy
372 results are in Sec. F.

Table 5: Synthetic ruler, ISIC 2018, DINOv2 ViT-B/14 @518: AUROC and reversed-test vs ERM [95% CI]

Method	Needs	clean@0	rev@0	clean@50	rev@50	clean@100	rev@100	rev@0	rev@50	rev@100
ERM		0.797	0.665	0.793	0.573	0.792	0.515			
ROI masking	ROI mask (train+test)	0.849	0.849	0.829	0.476	0.825	0.386	+0.184 [+0.146, +0.224]	-0.097 [-0.138, -0.056]	-0.129 [-0.170, -0.087]
Inpainting (oracle)	artifact pixels (train+test)	0.810	0.806	0.809	0.783	0.807	0.759	+0.141 [+0.120, +0.162]	+0.211 [+0.187, +0.234]	+0.243 [+0.218, +0.269]
Group-balanced	image-level A (train)	0.814	0.802	0.815	0.792	0.806	0.796	+0.137 [+0.109, +0.168]	+0.219 [+0.188, +0.252]	+0.281 [+0.248, +0.315]
DFR	image-level A (val)	0.786	0.789	0.787	0.794	0.789	0.794	+0.124 [+0.091, +0.160]	+0.221 [+0.184, +0.261]	+0.279 [+0.240, +0.320]
LEACE (paired, oracle renders)	paired renders (train)	0.829	0.852	0.810	0.810	0.809	0.809	+0.167 [+0.139, +0.197]	+0.257 [+0.212, +0.263]	+0.294 [+0.265, +0.325]
Repair consistency	artifact pixels (train)	0.828	0.707	0.822	0.616	0.814	0.558	+0.042 [+0.015, +0.069]	+0.043 [+0.014, +0.072]	+0.042 [+0.012, +0.073]
Insert-then-erase (artifact templates)	artifact templates	0.848	0.844	0.850	0.833	0.847	0.818	+0.179 [+0.152, +0.206]	+0.260 [+0.229, +0.291]	+0.303 [+0.271, +0.336]
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.841	0.842	0.843	0.840	0.844	0.842	+0.177 [+0.151, +0.205]	+0.267 [+0.238, +0.298]	+0.327 [+0.293, +0.362]
Insert-then-erase rank-1 (ablation)	artifact templates	0.811	0.799	0.809	0.770	0.806	0.730	+0.134 [+0.115, +0.154]	+0.197 [+0.174, +0.220]	+0.215 [+0.190, +0.240]
Insertion augmentation	artifact templates	0.814	0.726	0.812	0.654	0.810	0.603	+0.061 [+0.049, +0.074]	+0.081 [+0.068, +0.094]	+0.088 [+0.077, +0.099]

373 C.2 Mechanism: retention, not occlusion or context

374 Three explanations were tested separately. *Occlusion*: with the ruler present but uncorrelated with
375 the label (50/50), masking helps at every overlap (+0.036, +0.040, +0.039 at 0/50/100%); covering
376 lesion pixels costs nothing measurable, so the harm requires the correlation. *Lost context*: dilating
377 the mask at 50% overlap *increases* harm ($-0.097 \rightarrow -0.138 \rightarrow -0.154$ for margins 0/10/25 px) as
378 the margin restores ruler pixels (retention $0.50 \rightarrow 0.73 \rightarrow 0.95$); at 100% overlap, where retention is
379 already 1, harm is flat (-0.129 to -0.144) although the ruler’s share of visible pixels falls threefold.
380 *Amplified reliance*: masking doubles the same-head counterfactual sensitivity to the ruler ($|\Delta p|$
381 $0.235 \rightarrow 0.491$ at 100%). Harm is largest for small lesions (-0.364 vs $+0.064$ for large lesions at
382 100%), where the ruler is a larger share of what survives the mask.

383 C.3 Real hair (ISIC 2019), pre-registered

384 Two traps share one hair-free group (at most 30 hair pixels): Trap A contains hair lying mostly
385 inside the lesion ($r \geq 0.5$), Trap B hair mostly outside ($r < 0.1$); cells are matched to the
386 hair-free source mix, and five seeds of five-fold group-safe cross-validation give 172 reversed-test
387 melanomas per trap. Masking helps in Trap B (+0.107 [+0.078, +0.135]), harms in Trap A (-0.037
388 [$-0.057, -0.019$], negative in all five seeds) and the crossover is +0.145 [+0.105, +0.181]. The
389 harm holds in BCN images (-0.051 [$-0.075, -0.027$]); in HAM it is not distinguishable from
390 zero (-0.009 [$-0.036, +0.017$]). Oracle hair inpainting helps in both traps (+0.103, +0.096).
391 Group-balanced training and DFR, which need only an image-level hair label, give the largest gains
392 (+0.225/+0.251 in Trap A, +0.209/+0.215 in Trap B). Paired linear erasure recovers less than
393 half of that (+0.086/+0.081). The magnitude of the in-ROI harm depends on the definition of the
394 comparison group: with a looser hair-free group ($< 0.1\%$ of pixels) and only heavily haired images,
395 masking gives no benefit but no measurable harm inside the lesion (+0.012 [$-0.042, +0.062$]) while
396 still helping outside (+0.178); the benefit shrinks to zero as more of the hair lies inside the lesion.

397 With the dermatology foundation model DermLIP, the controlled ruler sweep reverses as with DI-
398 NOv2 (+0.161 at no overlap, -0.263 to -0.309 at 25–100%), but on real hair masking harms
399 at both locations (Trap A -0.085 [$-0.107, -0.063$], Trap B -0.088): with this encoder masking
400 removes disease context regardless of where the hair lies, so no location gap appears.

401 C.4 What the real-hair shortcut is made of

402 Removing hair pixels at test time only closes 44–47% of the ERM clean–reversed gap: most of
403 what a frozen encoder exploits in a real hair trap is not the hair pixels but what accompanies hair
404 (hair-bearing images differ in sex, age and body site). This bounds every pixel-level method, in-
405 cluding oracle inpainting and U-MtE, and explains why image-level-label methods, which remove
406 everything correlated with the label, recover more.

407 C.5 Reversed-environment gains can be gamed

408 Erasing the artifact concept from unpaired image-level labels yields the largest reversed gain in the
409 study (+0.326 in Trap A) while *inverting* the correlated/reversed ordering (0.618 vs 0.817). Post-
410 hoc prevalence-equalised recalibration [Kina and Petersen, 2026] behaves the same way under the
411 trap’s 90/10 shift (+0.427; correlated 0.433 vs reversed 0.918). Both remove diagnosis-correlated
412 signal or over-correct the group prior rather than removing reliance on the artifact; we therefore read
413 every reversed gain together with clean AUROC and the correlated/reversed ordering.

D Results: ultrasound, capsule endoscopy and chest radiography

D.1 Thyroid ultrasound: sonographer calipers

TN3K images with TNCD labels (3,493 images, 1,210 malignant) carry measurement calipers that a sonographer places on or around the nodule. Calipers were segmented by an audited rule-based detector (image-level precision and recall 0.86–0.89). With the E13 protocol unchanged, masking the nodule helps greatly when the calipers lie outside it (Trap B: +0.341 [+0.320, +0.363]) and much less when they lie inside it (Trap A: +0.111 [+0.097, +0.124]); the crossover is +0.230 [+0.204, +0.255] with DINOv2 and +0.163 [+0.132, +0.194] with MedSigLIP. After masking, the Trap A model still relies on the calipers (correlated 0.902 versus reversed 0.400). With synthetic calipers swept across the nodule boundary (1,663 caliper-free images), masking gains +0.402 [+0.347, +0.458] at no overlap and −0.030 [−0.079, +0.016] at full overlap (interaction +0.432 [+0.377, +0.492]): the entire benefit of masking disappears, without significant harm.

D.2 Capsule endoscopy: intestinal debris

SEE-AI frames (erosion 3,550 versus polyp-like 1,931; expert lesion boxes as ROI) with contamination masks from a patch-token probe trained on 2,160 expert-masked frames (held-out pixel accuracy 0.92, IoU 0.62). Leakage groups are contiguous blocks of 100 frame numbers plus perceptual-hash duplicates (no video identifiers are published). Masking helps against out-of-ROI debris (+0.465 [+0.440, +0.490]) and much less against debris on the lesion (+0.097), a crossover of +0.368 [+0.340, +0.397]. With synthetic debris on the 796 cleanest frames the reversal appears: +0.103 [+0.060, +0.149] at no overlap, −0.102 [−0.162, −0.040] at full overlap (interaction +0.205 [+0.150, +0.259]).

D.3 Chest radiography: a boundary case

With a synthetic tube swept into the lung fields of NIH pneumothorax radiographs (RAD-DINO), the encoder does not use an extra-pulmonary tube at all (reversed 0.913 versus clean 0.915 at no overlap), so masking has nothing to remove; inside the lungs the tube is used (reversed 0.614) and lung masking only partly removes it (0.725). On real devices (RANZCR-CLiP linked to NIH; central lines and endotracheal tubes) masking helped for both locations. The location-dependent *harm* of masking is therefore not universal; what is universal in our data is that masking leaves an in-ROI shortcut largely intact.

E Results: natural distribution, fine-tuning and label robustness

E.1 Unaltered test distributions

Without any resampling, in-ROI artifacts are naturally associated with the label in every cohort (thyroid calipers: 37.6% of benign vs 24.1% of malignant nodules; in-lesion hair: 31% of melanomas vs 20% of benign lesions; capsule debris on the lesion: 23% of polyp-like lesions vs 10% of erosions). We trained on natural training data and tested on untouched data: the official TN3K test split, each ISIC source held out in turn (a real change of hospital and device), and held-out capsule frame blocks. On the shortcut-conflicting pairs, masking *lowers* AUROC for thyroid (−0.102 [−0.139, −0.066]) and for two of the three held-out hospitals (BCN −0.016 [−0.027, −0.005], MSK −0.056 [−0.084, −0.029]); it helps for HAM (+0.053) and does not differ significantly from ERM on capsule. U-MtE with group balancing improves on masking for these pairs in every cohort (+0.014 to +0.088) and gives the best or tied-best cross-group AUROC in three of five settings (HAM 0.786, BCN 0.734 vs 0.735, capsule 0.947); plain group balancing is best on thyroid and MSK. Without artifact labels, U-MtE improves on masking for thyroid (+0.051) and HAM (+0.023) only.

Clinical secondary metrics Table 6 reports AUPRC, Brier score, expected calibration error and sensitivity/specificity at the operating point fixed on validation data. On the official thyroid split masking lowers sensitivity from 0.734 to 0.347 while specificity rises from 0.619 to 0.888 (balanced accuracy 0.677 → 0.618); U-MtE recovers part of the sensitivity (0.425) and has the best AUPRC

Table 6: Clinical secondary metrics on the unaltered test sets (Sec. E): mean (SD) over training seeds. Sensitivity and specificity at the operating point fixed on clean validation data; ECE with 10 equal-width bins; AUPRC baseline = prevalence.

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sens	Spec
Thyroid (official split) ($\pi = 0.38$)	ERM	0.736 (0.009)	0.606 (0.012)	0.207 (0.005)	0.082 (0.008)	0.734 (0.047)	0.619 (0.061)
	mask	0.731 (0.006)	0.632 (0.007)	0.214 (0.006)	0.120 (0.026)	0.347 (0.095)	0.888 (0.041)
	U-MtE	0.742 (0.003)	0.638 (0.007)	0.203 (0.001)	0.085 (0.007)	0.425 (0.088)	0.854 (0.050)
	balanced	0.716 (0.007)	0.584 (0.012)	0.219 (0.004)	0.110 (0.007)	0.714 (0.052)	0.599 (0.061)
	U-MtE _{bal}	0.731 (0.004)	0.630 (0.007)	0.207 (0.001)	0.084 (0.006)	0.399 (0.069)	0.866 (0.034)
ISIC, BCN held out ($\pi = 0.23$)	ERM	0.786 (0.003)	0.586 (0.003)	0.239 (0.006)	0.279 (0.011)	0.842 (0.038)	0.519 (0.075)
	mask	0.769 (0.004)	0.571 (0.009)	0.286 (0.007)	0.332 (0.009)	0.860 (0.013)	0.439 (0.023)
	U-MtE	0.773 (0.004)	0.574 (0.008)	0.280 (0.009)	0.328 (0.012)	0.858 (0.039)	0.448 (0.089)
	balanced	0.781 (0.005)	0.585 (0.009)	0.228 (0.006)	0.257 (0.011)	0.811 (0.046)	0.554 (0.087)
	U-MtE _{bal}	0.774 (0.004)	0.568 (0.008)	0.279 (0.011)	0.325 (0.016)	0.860 (0.027)	0.447 (0.051)
ISIC, MSK held out ($\pi = 0.19$)	ERM	0.770 (0.008)	0.477 (0.012)	0.172 (0.007)	0.161 (0.019)	0.608 (0.049)	0.786 (0.039)
	mask	0.790 (0.004)	0.512 (0.009)	0.146 (0.003)	0.115 (0.009)	0.593 (0.052)	0.824 (0.037)
	U-MtE	0.783 (0.004)	0.500 (0.008)	0.150 (0.003)	0.126 (0.009)	0.566 (0.073)	0.831 (0.049)
	balanced	0.766 (0.008)	0.462 (0.014)	0.178 (0.008)	0.170 (0.020)	0.626 (0.047)	0.755 (0.046)
	U-MtE _{bal}	0.774 (0.006)	0.482 (0.009)	0.154 (0.004)	0.131 (0.010)	0.594 (0.061)	0.798 (0.051)
ISIC, HAM held out ($\pi = 0.11$)	ERM	0.750 (0.005)	0.353 (0.005)	0.162 (0.006)	0.234 (0.011)	0.535 (0.092)	0.813 (0.068)
	mask	0.789 (0.002)	0.404 (0.006)	0.143 (0.005)	0.217 (0.011)	0.559 (0.112)	0.842 (0.063)
	U-MtE	0.799 (0.008)	0.406 (0.009)	0.145 (0.007)	0.226 (0.016)	0.573 (0.067)	0.846 (0.041)
	balanced	0.747 (0.017)	0.346 (0.019)	0.156 (0.014)	0.228 (0.015)	0.539 (0.078)	0.803 (0.076)
	U-MtE _{bal}	0.802 (0.005)	0.401 (0.009)	0.139 (0.006)	0.215 (0.021)	0.606 (0.054)	0.825 (0.037)
Capsule (held-out frames) ($\pi = 0.56$)	ERM	0.889 (0.019)	0.897 (0.046)	0.130 (0.016)	0.042 (0.011)	0.770 (0.062)	0.844 (0.055)
	mask	0.967 (0.011)	0.971 (0.014)	0.069 (0.013)	0.029 (0.011)	0.859 (0.032)	0.933 (0.025)
	U-MtE	0.962 (0.009)	0.967 (0.013)	0.081 (0.017)	0.065 (0.045)	0.871 (0.042)	0.897 (0.043)
	balanced	0.881 (0.024)	0.889 (0.053)	0.136 (0.017)	0.051 (0.009)	0.758 (0.087)	0.838 (0.059)
	U-MtE _{bal}	0.959 (0.008)	0.964 (0.012)	0.079 (0.007)	0.044 (0.012)	0.866 (0.048)	0.897 (0.052)

(0.638) and Brier score (0.203). For the held-out BCN hospital masking worsens both Brier score (0.239 $\rightarrow$ 0.286) and calibration (ECE 0.279 $\rightarrow$ 0.332); for HAM, MSK and capsule it improves every metric, in line with the AUROC results.

E.2 End-to-end fine-tuning

Fine-tuning ResNet-50 end-to-end (five folds) reproduces the frozen-backbone findings. Capsule: masking harms in-ROI ($-0.284 [-0.347, -0.226]$) and helps out-of-ROI ($+0.301$), crossover $+0.584 [+0.469, +0.691]$. Thyroid: the crossover points the same way ($+0.057 [-0.021, +0.130]$, five folds), and an end-to-end analogue of U-MtE—an invariance penalty between masked images with and without a generic overlay—improves on masking by $+0.143 [+0.099, +0.188]$ without shortcut flipping (reversed 0.679, correlated 0.703), matching the best label-using arm. It fails on capsule (-0.021), where the artifact resembles the pathology. Ovary (1,202 images) is underpowered: U-MtE end-to-end has the best reversed, correlated and clean AUROC, with CIs overlapping masking. A fine-tuned ViT-S reproduces the crossover on thyroid ($+0.168 [+0.074, +0.265]$), but the end-to-end U-MtE analogue does not improve on masking with this architecture ($+0.009 [-0.021, +0.040]$).

E.3 Zero-shot vision-language models

Without any task training, the foundation models’ own zero-shot scores (similarity to “benign” versus “malignant” text prompts) are already swayed by artifacts: on capsule frames the correlated and reversed AUROC differ by 0.40. Masking changes these zero-shot scores exactly as the location law predicts: for MedSigLIP it leaves reversed AUROC unchanged for in-ROI artifacts but improves it for out-of-ROI artifacts (thyroid $+0.013$ vs $+0.116$, crossover $+0.103 [+0.079, +0.127]$; capsule crossover $+0.049 [+0.024, +0.076]$), and it harms more for in-ROI artifacts on ovarian ultrasound (crossover $+0.181 [+0.135, +0.228]$) and for DermLIP on hair ($+0.051 [+0.031, +0.071]$). The location effect is therefore a property of the representation, not of the classifier trained on top of it.

Table 7: In-ROI artifacts (Trap A): reversed-test AUROC, with min(reversed, correlated) in parentheses (penalises shortcut flipping). Best robust value per column in bold. Generated by scripts/make_main_table.py.

Method	Needs	Hair/DINOv2	Hair/DeinLIP	Thyroid/DINOv2	Thyroid/MedSigLIP	Thyroid/ConcNcXt	Ovary/DINOv2	Ovary/MedSigLIP	Capsule/DINOv2	Capsule/MedSigLIP	Capsule/ConcNcXt	DeinRAD-DINO	Dein/DINOv2
ERM		0.49 (0.49)	0.67 (0.67)	0.29 (0.29)	0.28 (0.28)	0.35 (0.35)	0.41 (0.41)	0.42 (0.42)	0.59 (0.59)	0.59 (0.59)	0.55 (0.55)	0.19 (0.19)	0.28 (0.28)
ROI masking	ROI	0.45 (0.45)	0.58 (0.58)	0.40 (0.40)	0.35 (0.35)	0.36 (0.36)	0.55 (0.55)	0.43 (0.43)	0.68 (0.68)	0.70 (0.70)	0.64 (0.64)	0.24 (0.24)	0.29 (0.29)
Mask + overlay augmentation	ROI	0.46 (0.46)		0.42 (0.42)	0.38 (0.38)	0.37 (0.37)	0.53 (0.53)		0.70 (0.70)	0.72 (0.72)	0.65 (0.65)	0.24 (0.24)	0.29 (0.29)
U-MtE (ours)	ROI	0.51 (0.51)	0.51 (0.51)	0.62 (0.62)	0.55 (0.55)	0.49 (0.49)	0.56 (0.56)	0.64 (0.64)	0.57 (0.57)	0.47 (0.47)	0.56 (0.56)	0.27 (0.27)	0.28 (0.28)
U-MtE, protected (ours)	ROI + Y' of A=0	0.49 (0.49)	0.59 (0.59)	0.62 (0.62)	0.61 (0.61)	0.55 (0.55)	0.59 (0.59)	0.57 (0.57)	0.67 (0.67)	0.67 (0.67)	0.64 (0.64)	0.27 (0.27)	0.28 (0.28)
Group-balanced	A	0.72 (0.72)	0.81 (0.81)	0.68 (0.68)	0.70 (0.70)	0.66 (0.66)	0.71 (0.71)	0.73 (0.73)	0.78 (0.78)	0.81 (0.81)	0.72 (0.72)	0.54 (0.54)	0.40 (0.40)
DFR	A	0.74 (0.74)	0.78 (0.78)	0.74 (0.74)	0.76 (0.72)	0.71 (0.67)	0.72 (0.69)		0.80 (0.80)	0.80 (0.80)	0.77 (0.77)	0.68 (0.68)	0.55 (0.55)
Mask + DFR	ROI + A	0.69 (0.69)		0.83 (0.74)	0.81 (0.76)	0.77 (0.72)	0.75 (0.75)		0.88 (0.88)	0.90 (0.90)	0.86 (0.86)	0.65 (0.65)	0.57 (0.57)
U-MtE + balanced (ours)	ROI + A	0.70 (0.70)	0.65 (0.65)	0.79 (0.79)	0.79 (0.79)	0.74 (0.74)	0.74 (0.74)	0.79 (0.79)	0.86 (0.86)	0.85 (0.85)	0.80 (0.80)	0.46 (0.46)	0.44 (0.44)

E.4 Robustness to artifact-label definitions

Artifact masks for calipers come from a detector (visual audit: 95% of thyroid detections lie on real calipers; 92–97% of caliper-free images contain no visible caliper). Re-running the thyroid and ovary analyses with large-marker-only (≥ 50 px) and strict-location ($r \geq 0.7 / r < 0.05$) definitions leaves the crossover (thyroid +0.215 to +0.262; ovary +0.170 to +0.195) and the U-MtE_{protect} gain over masking (thyroid +0.217 to +0.282; ovary +0.043 to +0.067) significant in every variant; both grow under stricter definitions, as expected if detector errors dilute rather than create the effects.

E.5 Stricter leakage grouping

TN3K/TNCD, MMOTU and SEE-AI publish no patient or video identifiers, so the main analyses group near-duplicates by perceptual hash (and, for capsule, blocks of 100 consecutive frames). As a pre-registered sensitivity analysis we united these groups with connected components of mean-centred DINOv2 embeddings at the most aggressive cosine threshold that keeps every group below 5% of the cohort (thyroid: 3,455 $\rightarrow$ 3,081 groups, $\tau = 0.80$; ovary: 1,059 $\rightarrow$ 969, $\tau = 0.80$; capsule: 137 $\rightarrow$ 135, $\tau = 0.93$) and re-ran all primary contrasts. Eleven of twelve keep their sign and significance: the location crossover is +0.235 [+0.211, +0.259] (thyroid), +0.157 [+0.080, +0.235] (ovary) and +0.394 [+0.359, +0.430] (capsule), and protected U-MtE still beats masking on thyroid (+0.251 [+0.219, +0.279]) and ovary (+0.047 [+0.020, +0.072]). The exception is the small ovary advantage of erasure over augmentation, which loses significance (+0.020 [−0.011, +0.053]). Embedding grouping removes visually near-identical images of one patient or session; it cannot exclude differently looking images of the same patient.

F Results: artifact-agnostic mask-then-erase

F.1 U-MtE repairs what masking leaves behind

Without any artifact example, artifact mask or artifact label—it uses only the ROI mask that masking already requires—U-MtE improves reversed AUROC over masking in the in-ROI regime of every modality: thyroid calipers +0.222 [+0.197, +0.251] (DINOv2) and +0.204 [+0.174, +0.237] (MedSigLIP); dermoscopic hair +0.052 [+0.044, +0.062]; synthetic capsule debris +0.081 [+0.050, +0.112]; synthetic calipers +0.278 [+0.232, +0.325]; synthetic chest tubes +0.134 [+0.120, +0.151] (DINOv2 / RAD-DINO). With MedSigLIP the unbalanced method *loses* to masking in both controlled sweeps (debris −0.114 [−0.147, −0.082], calipers −0.076 [−0.110, −0.043]) although it wins on real thyroid calipers (+0.204); the balanced variant beats masking in all four controlled sweeps (+0.193 to +0.471) and has no location interaction in any of them. Disease protection turns the MedSigLIP debris failure into a gain (+0.061 [+0.032, +0.092] over masking) but not the caliper one (−0.041 [−0.089, −0.005]).

F.2 Erasure, not augmentation

Using the same generic overlays as training augmentation (overlaid copies of a random half of the training images, in the masked view) gains at most +0.03 over masking; erasing their subspace instead gains +0.207 [+0.182, +0.235] (thyroid, DINOv2), +0.177 [+0.148, +0.211] (thyroid, MedSigLIP) and +0.045 [+0.037, +0.054] (hair) more than augmentation. Under the stricter

near-duplicate grouping (Sec. E.5) the thyroid advantage is unchanged (+0.192) but the small ovary advantage (+0.035) loses significance, so we regard erase > augment as established for thyroid and hair only.

F.3 When the artifact resembles the disease

On real capsule debris U-MtE is worse than masking (−0.115): erosions carry yellow-white fibrin that resembles debris, and a subspace estimated from real-debris templates is worse still (−0.395). Disease protection removes the failure (−0.010 [−0.020, −0.001] generic, −0.012 [−0.028, +0.003] template) at a cost of at most 0.012 where erasure already works (thyroid −0.005 [−0.036, +0.024], hair −0.012 [−0.020, −0.003]).

F.4 With image-level artifact labels

Adding group balancing, U-MtE_{bal} is the most robust model in thyroid ultrasound by the minimum of correlated and reversed AUROC (0.788 DINOv2, 0.792 MedSigLIP; DFR 0.659 and 0.719); in real capsule endoscopy masking plus last-layer retraining is higher (0.885 versus 0.857); on dermoscopic hair DFR alone is best (0.741) because most of the hair shortcut is not in the hair pixels (removing every hair patch with the oracle mask gains only +0.015). In both controlled sweeps the benefit of U-MtE_{bal} no longer depends on location (interaction −0.000 [−0.023, +0.023] capsule; −0.054 [−0.098, −0.004] thyroid), whereas that of masking does (+0.205, +0.432). The capsule findings replicate with MedSigLIP (crossover +0.332 [+0.296, +0.365]; masking plus DFR 0.901). All thyroid and capsule findings replicate with a CNN (ConvNeXt-Base): crossovers +0.284 and +0.292; U-MtE − mask +0.128 [+0.112, +0.145] and erase − augment +0.120 [+0.102, +0.139] on thyroid.

F.5 Label-free and erasure baselines

Among methods that need no artifact labels, U-MtE beats JTT [Liu et al., 2021] by +0.198 [+0.167, +0.231] (thyroid), +0.137 [+0.098, +0.179] (ovary) and +0.094 [+0.077, +0.109] (chest drains, RAD-DINO; not with DINOv2, −0.010); JTT on the unmasked view is best on hair. Task-preserving concept removal (SPLINCE; Holstege et al., 2025), fitted with artifact labels in the whitened, minimal-distortion form of the original method, removes the artifact covariance at a large cost in disease signal (clean AUROC 0.56–0.78 versus 0.69–0.87 for masking) and still flips the shortcut on capsule endoscopy (reversed 0.868, correlated 0.683): when artifact and label are strongly associated in training, removing one removes much of the other. By min(reversed, correlated) U-MtE_{bal} exceeds SPLINCE on thyroid, capsule and hair (0.788 vs 0.596; 0.857 vs 0.683; 0.704 vs 0.564); on chest drains SPLINCE is higher (0.609 vs 0.460) but below DFR (0.683).

F.6 Text-prompted erasure

With vision–language backbones an artifact direction can be obtained from text alone (“a thyroid ultrasound image with measurement calipers” minus “a thyroid ultrasound image”; biased-prompt construction, Chuang et al., 2023). Erasing that subspace after masking is a near no-op in every cohort (+0.001 to +0.008 over masking; MedSigLIP thyroid, ovary, capsule and DermLIP hair), whereas U-MtE with the same MedSigLIP features gains +0.204 [+0.174, +0.237] (thyroid) and +0.215 [+0.182, +0.246] (ovary). The direction along which a described artifact moves the *text* embedding is not the direction along which the real artifact moves the *image* embedding; synthetic insertions measure the latter directly.

F.7 A single procedure: validation-driven selection

Selecting, per fold, the candidate with the best group-aware validation score (image-level artifact labels needed on validation images only) beats masking in all five cohorts (+0.17 to +0.40), is within 0.02 of the best candidate in each, and beats DFR in three (thyroid +0.057, capsule +0.047, ovary +0.010); it chooses U-MtE_{bal} where the artifact is an overlay and plain balancing on hair and drains, where DFR remains better. A split-validation variant that admits DFR as a candidate is reported in the supplement.

F.8 Modern inpainting

Inpainting the detected caliper pixels with LaMa [Suvorov et al., 2022] and then masking is a strong baseline when artifact masks are available (thyroid +0.226 [+0.213, +0.238], ovary +0.123 [+0.089, +0.155] over masking). U-MtE_{protect}, which needs no artifact mask, matches it on thyroid (−0.009 [−0.045, +0.023]) and is below it on ovary (−0.080) and on dermoscopic hair, where published hair masks are available (−0.086); with image-level artifact labels, U-MtE_{bal} exceeds it in all three (+0.162, +0.067, +0.123).

G Statistical plan and pre-registration

Every experiment was specified in a dated, publicly versioned document before any model was fitted on its data. The 16 primary tests (location crossover, protected U-MtE vs masking, erase vs augment, U-MtE_{bal} vs balanced; four cohorts each) are Holm-corrected; 12/16 are supported (Fig. 4). Everything else is secondary. Datasets considered and not used (CANDID-PTX: no access; breast ultrasound: no artifact-free group; colonoscopy: no reflection-free images; ISIC ink, cardiomegaly lines: too few images; pathology pen marks: unreliable masks) and the leakage analysis are documented in the repository.

H Compute

All experiments ran on one RTX 3090 (24 GB), 8 CPU cores and 31 GB RAM; the full pipeline takes about two days, dominated by feature extraction and fine-tuning.

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